

MINERAL WELLS, TX, USA

WMO: 722597

Lat: 32.782N

Lon: 98.060W

Elev: 284

StdP: 97.97

Time Zone: -6.00 (NAC)

Period: 00-19

WBAN: 93985

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-6.3	-4.1	-13.5	1.2	0.1	-11.5	1.5	2.8	11.5	9.6	10.6	10.7	3.3	350	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.7	38.9	22.8	37.4	23.0	36.1	23.0	25.3	33.0	24.9	32.5	24.3	32.0	4.4	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.3	18.7	27.9	22.8	18.1	27.4	22.3	17.6	27.1	78.9	33.0	76.7	32.6	74.8	32.1	29.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.2	8.9	8.1	DB	-10.0	40.6	2.2	1.7	-11.5	41.9	-12.8	42.9	-14.0	43.9	-15.6	45.1	(4)
(5)			WB	-10.7	26.5	2.1	1.1	-12.2	27.3	-13.4	27.9	-14.6	28.5	-16.1	29.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.6	7.4	9.3	14.3	18.6	22.6	27.1	28.8	28.9	25.1	19.0	13.2	8.1	(6)	
	DBStd	8.89	5.39	6.13	5.41	4.42	3.74	2.53	2.28	2.48	3.19	4.74	5.34	5.38	(7)	
	HDD10.0	353	116	79	22	3	0	0	0	0	0	3	28	102	(8)	
	HDD18.3	1336	340	257	146	48	10	0	0	0	1	50	165	319	(9)	
	CDD10.0	3488	34	61	156	261	391	512	584	587	452	282	126	43	(10)	
	CDD18.3	1429	1	4	21	57	143	262	326	329	202	71	12	2	(11)	
	CDH23.3	16619	24	67	231	574	1403	3046	4143	4150	2109	722	125	24	(12)	
	CDH26.7	8509	2	15	62	188	564	1593	2380	2393	1017	266	27	3	(13)	
(14) Wind	WSAvg	3.9	3.8	4.2	4.5	4.7	4.3	3.9	3.5	3.2	3.1	3.6	3.8	3.8	(14)	
(15) Precipitation	PrecAvg	817	43	54	73	76	99	81	54	58	76	79	67	46	(15)	
	PrecMax	1287	131	214	189	268	232	275	131	170	184	202	231	222	(16)	
	PrecMin	443	2	5	0	7	40	8	0	0	3	6	1	0	(17)	
	PrecStd	213	30	49	51	55	53	68	42	38	47	60	59	44	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	28.8	31.1	34.1	35.5	39.0	40.8	41.1	37.4	33.6	29.6	26.3	(19)	
		MCWB	14.7	16.4	17.5	20.3	22.5	22.8	22.4	23.0	22.1	21.5	18.7	16.7	(20)	
	2%	DB	22.8	25.3	28.4	31.0	33.6	36.9	38.9	38.9	35.3	31.7	26.6	22.8	(21)	
		MCWB	13.6	15.1	17.4	19.3	22.4	22.8	22.7	23.0	22.6	21.0	17.7	14.2	(22)	
	5%	DB	19.8	22.4	25.9	28.7	31.8	35.3	37.3	37.5	33.8	29.7	24.3	20.2	(23)	
		MCWB	12.5	14.1	16.4	19.1	21.7	23.0	23.1	23.1	22.4	19.9	16.5	13.3	(24)	
	10%	DB	17.3	19.5	23.4	26.6	30.0	33.8	35.8	36.2	32.3	27.7	22.0	17.3	(25)	
		MCWB	10.8	13.1	15.7	18.3	21.6	23.1	23.2	23.3	22.0	18.8	15.6	11.7	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	19.1	20.6	22.9	25.2	26.2	25.8	25.7	25.1	23.7	21.1	18.8	(27)	
		MCDB	21.5	24.2	26.8	28.8	32.4	33.0	33.3	34.4	32.5	29.7	25.9	22.5	(28)	
	2%	WB	16.0	17.7	19.3	21.8	24.1	25.2	25.2	25.0	24.2	22.8	19.4	16.9	(29)	
		MCDB	19.8	21.7	24.7	27.4	30.1	32.1	33.1	33.6	31.3	28.6	23.3	20.7	(30)	
	5%	WB	13.7	16.3	18.4	20.8	23.3	24.5	24.7	24.5	23.7	21.9	18.2	14.8	(31)	
		MCDB	18.1	20.4	23.5	25.8	29.4	31.4	32.6	33.0	30.3	26.9	22.2	18.2	(32)	
	10%	WB	11.4	14.3	17.2	20.0	22.6	23.9	24.1	24.0	23.1	20.9	17.0	12.4	(33)	
		MCDB	16.5	18.6	21.9	24.5	28.3	30.8	32.0	32.3	29.1	24.9	21.0	16.6	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	13.1	13.4	13.2	12.0	12.1	12.7	12.9	12.5	13.5	13.3	12.9	(35)	
		MCDBR	18.1	17.6	16.3	15.9	14.3	14.1	15.0	14.9	14.1	15.3	15.7	17.1	(36)	
	5% WB	MCWBR	10.4	9.4	7.7	7.2	5.7	4.0	3.7	3.6	4.5	6.6	8.0	9.6	(37)	
		MCWBR	15.0	14.5	12.6	12.0	11.9	11.7	12.1	12.8	11.6	12.2	12.3	13.0	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.304	0.314	0.344	0.382	0.416	0.425	0.429	0.421	0.393	0.352	0.327	0.309	(40)	
		taud	2.519	2.499	2.433	2.348	2.320	2.329	2.361	2.359	2.388	2.487	2.491	2.507	(41)	
	Ebn at Noon		920	944	935	907	874	862	856	860	872	888	883	892	(42)	
		Edh at Noon	84	95	110	125	130	128	123	122	114	95	86	81	(43)	
(44) All-Sky Solar Radiation	RadAvg		3.01	3.58	4.70	5.68	6.07	6.78	6.85	6.23	5.23	4.12	3.17	2.64	(44)	
	RadStd		0.31	0.55	0.34	0.47	0.61	0.60	0.45	0.40	0.43	0.62	0.41	0.33	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (47 neighbors)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page